

Shengyi “Costa” Huang

SUMMARY

I am a Member of Technical Staff at Periodic Labs, a startup that builds AI scientists to accelerate materials science. Previously, I was a Machine Learning Engineer at the Allen Institute for Artificial Intelligence, where I led post-training engineering efforts for state-of-the-art open-source language models, including Tülu 3 and OLMo 2. I hold a Ph.D. in Computer Science from Drexel University, specializing in deep reinforcement learning under the guidance of [Santiago Ontañón](#). I am the creator of [CleanRL](#), a top 20 most-starred deep RL library on GitHub.

EDUCATION

Ph.D. in Computer Science Sep 2023
M.S. in Computer Science Dec 2021

Drexel University, Philadelphia, PA

B.S in Computer Science, B.S in Mathematics May 2018
Furman University, Greenville, SC

EXPERIENCE

Member of Technical Staff May 2025 – Present
Periodic Labs, Inc., CA

Machine Learning Engineer Jul 2024 – May 2025
Allen Institute for Artificial Intelligence, Philadelphia, PA

- Spearheaded post-training engineering efforts for Tülu 3 and OLMo 2, establishing them among the leading open-source large language models in scientific research.
- Co-pioneered Reinforcement Learning from Verifiable Rewards (RLVR) as the final training stage for language models, significantly improving accuracy on mathematical reasoning and instruction-following tasks.
- Architected and implemented RLVR infrastructure from the ground up, successfully scaling model training to 405B parameter models, and supporting agentic capabilities such as writing code and tool use.

Machine Learning Engineer Jun 2023 – Jul 2024
Hugging Face, Philadelphia, PA

- Drove research and engineering deep dive in reinforcement learning from human feedback (RLHF), documenting and validating over 21 useful RLHF implementation details.
- Improved the training stability in multi-GPU / multi-node settings for Hugging Face’s highly popular open-source RLHF library TRL (6.6 GitHub stars, used by 700+ open repositories).
- Led the research on alignment techniques such as Anthropic’s Constitutional AI and support the development of Hugging Face’s Zephyr 7B model.

Data Scientist Intern Jun 2022 – Sep 2022
Riot Games, Los Angeles, CA

- Iterated deep reinforcement learning infrastructure to be used in the next generation of games.
- Conducted research at the AI Accelerator team, with a focus on applying cutting-edge multi-agent deep reinforcement learning algorithms to the next generation of games.
- Led the technical discussions on algorithm design and iterated hundreds of research experiments to create playful and intelligent game-playing agents.
- Implemented an automated pipeline that saved 80% human labor spent on the evaluation process.

Deep Learning Algorithm Engineering Intern

NVIDIA, Santa Clara, CA

Mar 2022 – Jun 2022

- Built better utilities in NVIDIA's Isaac Gym robotic simulation environment, which helps researchers to develop, test, and manage AI-based robots.
- Analyzed and benchmarked the performance of state-of-the-art deep RL frameworks such as Stable-baselines 3, rl_games, RLLib, and CleanRL, creating solid baselines for the research team.
- Implemented a more efficient multi-GPU training paradigm for rl_games that increased up to 20% more system throughput for training agents.

Deep Learning Growth Engineer Intern

Weights and Biases, Palo Alto, CA

Jun 2021 – Sep 2021

- Researched 37 implementation details of Proximal Policy Optimization. Published at ICLR 2022.
- Contributed the W&B visualization integration to popular RL projects, such as the MineRL competition (16k views) and Stable-baselines 3 library (used by 400+ packages).
- Created multiple ML educational content, including a blog post on AWS SageMaker and a webinar on experiment tracking and analysis workflow (200 + views on YouTube).

Machine Learning Engineer Intern

Curai Health, Palo Alto, CA

Apr 2021 – Jun 2021

Graduate Research / Teaching Assistant

Drexel University, Philadelphia, PA

Sep 2018 – Jun 2023

PROJECTS

CleanRL (github.com/vwxyzjn/cleanrl, 3.2k stars on GitHub)

High-quality single file implementation of Deep Reinforcement

Learning algorithms with research-friendly features

- Python PyTorch OpenAI Gym Tensorboard Docker AWS
Weights and Biases Deep Q-learning Policy Gradient Visualization

Gym-MicroRTS (github.com/vwxyzjn/gym-microrts, 100 stars on GitHub)

The OpenAI Gym wrapper of MicroRTS for deep RL research

- Python OpenAI Gym Policy Gradient Real-time Strategy Games
Docker AWS Learning through Self-play CI/CD Numpy

Portwarden (github.com/vwxyzjn/portwarden, 372 stars on GitHub)

Create Encrypted Backups of Your Bitwarden Vault with Attachments

- Go Docker Kubernetes AES Encryption

PUBLICATIONS

OLMo, T., Walsh, P., Soldaini, L., Groeneveld, D., Lo, K., Arora, S., Bhagia, A., Gu, Y., **Huang, S.**, Jordan, M., and others. (2024). 2 OLMo 2 Furious. In submission for COLM 2025.

Lambert, N., Morrison, J., Pyatkin, V., **Huang, S.**, Ivison, H., Brahman, F., Miranda, L. J. V., Liu, A., Dziri, N., Lyu, S., and others. (2024). Tülu 3: Pushing frontiers in open language model post-training. In submission for COLM 2025.

Noukhovitch, M., **Huang, S.**, Khonneux, S., Hosseini, A., Agarwal, R., and Courville, A. (2024). Asynchronous RLHF: Faster and More Efficient Off-Policy RL for Language Models. ICLR 2025.

Huang, S., Noukhovitch, M., Hosseini, A., Rasul, K., Wang, W., and Tunstall, L. The N+ Implementation Details of RLHF with PPO: A Case Study on TL; DR Summarization. COLM 2024.

Tunstall, L., Beeching, E., Lambert, N., Rajani, N., Rasul, K., Belkada, **Huang, S.**, Werra, L., Fourier, C., Habib, N., Sarrazin, N., Sanseviero, O., Rush, A., and Wolf, T. Zephyr: Direct distillation of lm alignment. COLM 2024.

Huang, S., Weng, J., Charakorn, R., Lin, M., Xu, Z., and Ontañón, S. Cleanba: A Reproducible and Efficient Distributed Reinforcement Learning Platform. *ICLR* 2023.

Sullivan, R., Kumar, A., **Huang, S.**, Dickerson P J., Suarez J., Reward Scale Robustness for Proximal Policy Optimization via DreamerV3 Tricks, *NeurIPS* 2023.

Weng, J., Lin, M., **Huang, S.**, Liu, B., Makoviichuk, D., Makoviyshuk, V., Liu, Z., Song, Y., Luo, T., Jiang, Y. and Xu, Z., EnvPool: A Highly Parallel Reinforcement Learning Environment Execution Engine, *NeurIPS* 2022.

Huang, S., Kanervisto, A., Raffin, A., Wang, W., Ontan'ón, S., & Dossa, R.F. A2C is a special case of PPO. *preprint, 2022*

Huang, S., Dossa, R., Raffin, A., Kanervisto, A., Wang, W. The 37 Implementation Details of Proximal Policy Optimization. *ICLR Blog Post Track, 2022*

Huang, S., Dossa, R., Ye, C., Braga, J., CleanRL: High-quality Single-file Implementations of Deep Reinforcement Learning Algorithms, *Journal of Machine Learning Research, 2022*

Huang, S., Ontañón, S., “A Closer Look at Invalid Action Masking in Policy Gradient Algorithms”, *FLAIRS-35, 2022*

Compton, R., Valmianski, I., Deng, L., **Huang, C.**, Katariya, N., Amatriain, X., Kannan, A. MED-COD: A Medically-Accurate, Emotive, Diverse, and Controllable Dialog System. *Machine Learning for Health, 2021.*

Dossa, R., **Huang, S.**, Ontañón, S., Matsubara, T., “An Empirical Investigation of Early Stopping Optimizations in Proximal Policy Optimization”, *IEEE Access, 2021*

Huang, S., Ontañón, S., Bamford, C., Grela, L., “Gym- μ RTS: Toward Affordable Full Game Real-time Strategy Games Research with Deep Reinforcement Learning”, *IEEE Conference on Games 2021*

Huang, S., Ontañón, S., “Measuring Generalization of Deep Reinforcement Learning Applied to Real-time Strategy Games”, *AAAI 2021 Reinforcement Learning in Games Workshop*

Bamford, C., **Huang, S.**, Lucas, S., “Griddly: A platform for AI research in games.”, *AAAI 2021 Reinforcement Learning in Games Workshop*

Huang, S., Ontañón, S., “Action Guidance: Getting the Best of Training Agents with Sparse Rewards and Shaped Rewards”, *AIIDE 2020 Strategy Games Workshop*

Huang, S., Ontañón, S., “Comparing Observation and Action Representations for Reinforcement Learning in μ RTS”, *AAIIDE 2019 Strategy Games Workshop*

Huang, S., Healy, C., “StreetTraffic: a Library for Traffic Flow Data Collection and Analysis”, poster presentation in *ACMSE 2018 Conference*

SERVICES

I was a reviewer for the following journals and conferences: COLM (2025), IJCAI (2024), ICLR (2024), NeurIPS (2023), NeurIPS (2022), JMLR (2022), JOSS (2022), ICGA(2021),

TALKS

12/2023, NeurIPS Neural MMO Workshop. Topics: CleanRL, Cleanba.

9/2024, InstaDeep. Topics: CleanRL, Open RL Benchmark.

9/2024, Cohere for AI. Topics: Cleanba.

1/2022, Riot Games, Topics: RL, Gym- μ RTS